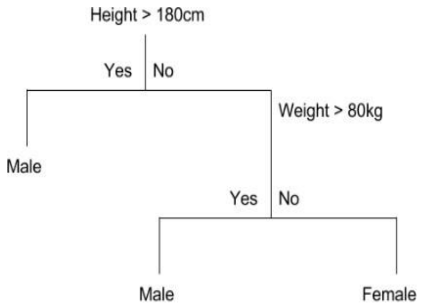
Decision Trees

Classification Algorithms

Decision Trees

- Most popular tool - Kaggle
- Easy to understand
- Like a flow chart
- if-else rules

Decision Trees




```
if (Height > 180cm)
        return Male
else
        if (Weight > 80kg)
                return Male
         else
                return Female
```

Pseudo Code for Decision Tree

- Place the best features of the dataset at the root of the tree
- Split the training set into subsets
- Repeat step 1 and step 2 on each subset until you find leaf nodes in all the branches of the tree

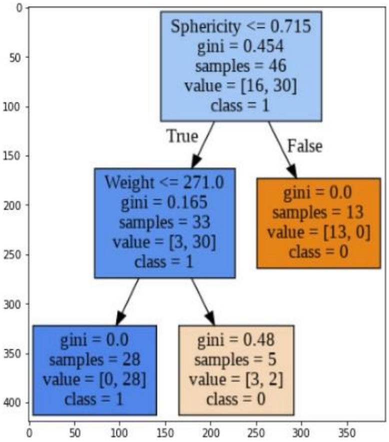
Decision Tree

- Splitting Criterion 1: Based on Weight?
- Splitting Criterion 2: Based on Height?
- Who chose the order?
- On what basis?

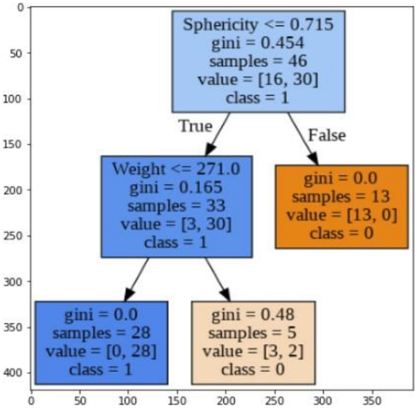
Experiment

- Demo_DT_Fruits_data
  - Dataset : Fruits
  - Objective : To classify data using Decision tree classifier

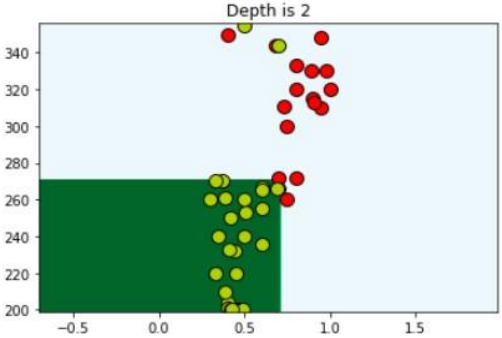
Decision Tree with Depth 2



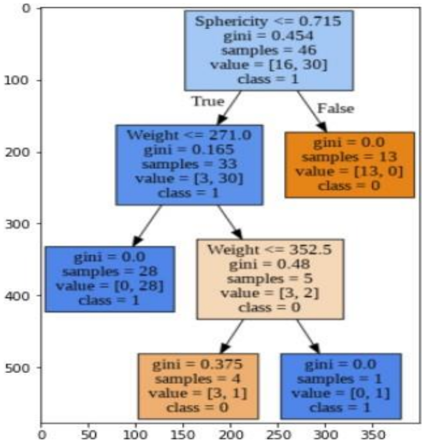
Decision Tree with Depth 2



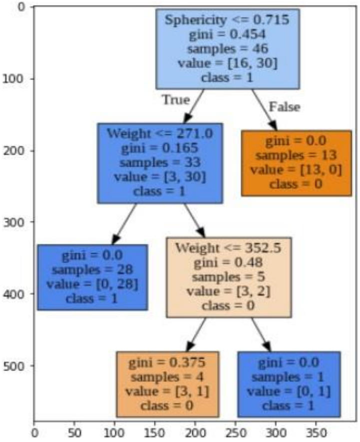
Training Accuracy: 0.96 Testing Accuracy: 0.83

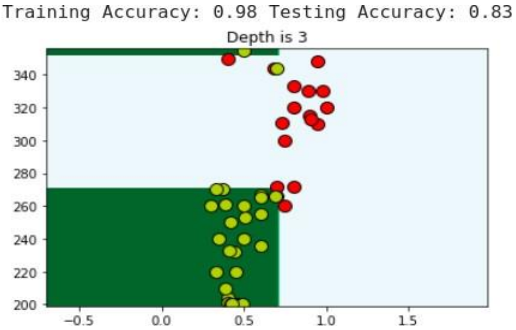


Decision Tree with Depth 3



Decision Tree with Depth 3





Summary

- Decision Tree:
  - Simple Human understandable solution
- Training: Finding the best tree
  - Best criteria
  - Best value to test the criteria against
  - Best sequence of criteria
- Testing: Use criteria obtained in training stage to classify

Thanks!!

Questions?